

## Chemistry for Materials/Polymer Chemistry

Comments: some questions in the question set are not like exam or test questions. They are used for the purpose of reflecting on some topic to ensure students understand the main concepts. These will be called here “Reflecting questions” (RQ). The questions more similar to exam/test questions will be labelled ETQ.

### Question Set 1

- 1) Have a look at the Introductory Lecture Notes Set 1 and ask questions of what you do not understand.
- 2) What are the similarities and the differences between  $M_n$ ,  $M_w$  and MW?

RQ/ETQ: They all refer to molecular weight.  $M_n$  is *number average molecular weight* used in polymer science.  $M_w$  is *weight average molecular weight* used in polymer science. MW is simply “molecular weight” and used when we are not being explicit on whether we refer to  $M_n$  or  $M_w$

- 3) In QMPLUS you can find an Excel spreadsheet called “Molecular weight calculations file”. This file contains two sheets: one is called MW. Open this sheet. It contains molecular weight data of a hypothetical polymer sample. Calculate the  $M_n$  and the  $M_w$  of the sample. Calculate the PDI of the sample. Be prepared to share your answers for discussion.

ETQ: this question is answered using the equations for  $M_n$  and  $M_w$  found in the Lecture Notes.

$$\overline{M}_n = \frac{\sum N_i M_i}{\sum N_i} \quad \overline{M}_w = \frac{\sum w_i M_i}{\sum w_i} = \frac{\sum N_i M_i^2}{\sum N_i M_i}$$
$$PDI = \frac{\overline{M}_w}{\overline{M}_n}$$

Using the  
equations:

|       |          |
|-------|----------|
| $M_n$ | 67187.5  |
| $M_w$ | 82465.12 |
| PDI   | 1.227388 |

- 4) In the same file, open the other sheet, called "Capitals". It has population data for all the Chinese provincial (or municipal) capital cities.

Please make sure you use the latest file with all the provincial/municipal capital cities of China.

- a. Use these data to calculate the population average of Chinese Provincial capitals (this is analogous to  $M_n$  for polymers).

RQ/ETQ: use the equation for  $M_n$ , substitute MW ( $M_i$ ) for population.  $N_i$  is always = 1 (there is only 1 Beijing, 1 Shanghai, 1 Xi'an, etc). " $M_n$ "=8.2million

- b. Now calculate a "weighted population average" (analogous to  $M_w$  for polymers). Of course, there is only one of each city, therefore,  $N_i$  is 1, for all the cases.

RQ/ETQ: use the equation for  $M_n$ , substitute MW ( $M_i$ ) for population.  $N_i$  is always = 1 (there is only 1 Beijing, 1 Shanghai, 1 Xi'an, etc). " $M_w$ "=12.7million

- c. What is the percentage of population living in any provincial capital city in China with less than 3M people? Be prepared to share your answers for discussion.

RQ: 3.6% (Total "Wi" living in cities of <3M population, for a total "Ni" 18% (% number of cities with <3M population). This shows there is a lot more people living in "bigger" cities and help students understand the need for a concept like  $M_w$ : what's important for people in a city is how many people live in the city with them, not the fact that they live "in a city". What's important for people in a city, is how big that city is.

For a polymer, the equivalent of 1 person is 1 repeating unit. Equally, what's important for 1 repeating unit is not that they belong in 1 molecule, but how big that molecule is.  $M_w$  reflects the size (mass) of the molecule to show this effect.

- 5) Imagine that Xian formed a hypothetical "Xian municipal administration", with 4 districts. These hypothetical 4 districts are Xian City (population: 8 million), Dongda I (population: 20,000), Dongda II (population: 30,000) and Dongda III (population 25,000).

RQ: similar to 4) but now a lot more unrealistic to emphasise the issue.

- a. Calculate the average population in the 4 districts of the hypothetical Xian municipal administration (analogous to  $M_n$  for polymers).

Using the equation,  $M_n=2M$  ( $N_i=1$ )

- b. Now calculate the "weighted average population" (analogous to  $M_w$  for polymers).

Using the equation,  $M_w=7.9M$  ( $N_i=1$ )

- c. What is the percentage of population living in "Xian municipal administration", who do not live in Xian City?

Less than 1% (0.92%)

- d. Based on the differences between the two type of calculations in a and b above, why do you think it is important to calculate population averages in different ways? Be prepared to share your answers for discussion.

You can see how different the values of  $M_n$  and  $M_w$  are. This is because in  $M_n$  all "towns" (molecules) count the same, no matter how big they are. So, 99% of the population considered live in a city of 8M people but calculating a normal average ( $M_n$ ) results in stating they live in a population of 2M. This isn't right. Therefore, another more realistic

value needs to be calculated. For polymers, you will rarely have this unrealistically large difference, but hopefully, you now understand the need to use  $M_w$  AND  $M_n$  to have an idea of polymer MW.

- e. Reflect on why it is important to have two types of calculations of MW for polymers.  
Done above in d).